What Does a Game-Trained Embedding Learn? Strategy Ensembles versus Latent Overhead in Pursuit–Evasion

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Abstract

Multi-head physics-informed networks learn low-dimensional embeddings of PDE solution families. We train one such network on a two-player pursuit–evasion game on the torus, a strategic rather than a dynamical problem, and ask what its latent space encodes. The key is to separate two objects. The *strategy ensemble* — the control curves the heads output, one per initial condition — is low-dimensional, as saddle-point analysis predicts: its participation ratio is 1.5–3.0, close to the ideal value of 2, and as compressible as the Burgers family the architecture was designed for. This rank is learned, not built in: game training collapses a planted rank-9 ensemble to rank 2–3. The *shared latent curve* carries a larger overhead ($\mathrm{PR} = 5.4 \pm 0.9$) that no head uses. Three controls trace this overhead to the input featurisation rather than the game: its spectrum breaks at the input Fourier bandwidth, it is uncorrelated with the game's winding classes across eight runs, and its variance is set mostly by the random initialisation ($4.7 : 1$ over the task instance). The two dimensionalities stay separate in all 43 training runs and across a sweep of the speed difference that carries the game through its capturability transition. The game's topology does appear — but as clusters of the strategy ensemble ($6\times$ separation between winding classes), not as latent dimensions. We are also careful about one tempting reading: the apparent asymmetry between the two players' spectra is partly an artifact of head gain, and what survives that is unstable under symmetric training, so we do not attribute it to the game's inf–sup structure. Every claim has a matching control, and an independently trained Hamilton–Jacobi–Isaacs critic checks the actor at the value level; its own failure modes are the subject of a companion paper.

Keywords: game-trained embeddings, representation learning, pursuit–evasion differential games, physics-informed neural networks, participation ratio, strategy ensembles, interpretability

1 Introduction

When a physics-informed network embeds a family of PDE solutions, its latent dimensions usually have a clear meaning: they parameterise the family. Tarancón-Álvarez et al. (2026) show this for the Burgers equation with a multi-head network — a shared body maps

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the base coordinates to a latent curve, and per-instance linear heads pick out solutions — with 90% of the family's variance in 3 of 20 latent components. It is tempting to reuse the architecture on a harder problem and read its spectrum the same way: train it on a two-player game and call the spectrum the dimensionality of strategic behaviour.

We try exactly that, and the tempting reading fails — while a sharper one holds up. The problem is a pursuit–evasion game on the flat torus (Isaacs, 1965; Başar and Olsder, 1999): a pursuer 20% faster than the evader over a fixed horizon, both built as head banks on shared bodies and trained by self-play on a differentiable capture-time loss. For this geometry, differential-game theory already says what the optimal open-loop strategies are — constant headings along the shortest displacement geodesic (Evans and Souganidis, 1984) — so it predicts, before any training, the dimension of what the network is meant to learn. That prediction is about the *strategy ensemble*: the control curves the network outputs, one per initial condition. It says nothing about the latent curve, which sits upstream of the heads and can carry anything.

The multi-head architecture is not new here. We take it from prior work. What is new is what we turn it on: a strategy embedding learned purely from a game, and a way to separate the embedding's strategic content from its overhead. Our method is to define those two objects, then attribute every observable to game structure, architecture, or training protocol with explicit controls. There are four main findings.

Two dimensionalities separate (Section 3). The strategy ensemble has participation ratio 1.5–3.0 across every protocol variant we ran. This matches the constant-heading analysis, and the ensemble is as compressible as the Burgers family (90% of variance in 4 of 32 ensemble components, against 3 of 20). A warm-start control shows the rank is actively selected rather than an architecture default: game training destroys a pre-fitted ensemble of rank 9.3 within 8,000 iterations. The latent curve keeps $\mathrm{PR} = 5.4 \pm 0.9$, an overhead that no head uses (stacked-head effective rank ≈ 2.5). The separation is not a population accident. In each of the 43 training runs and snapshots we produced, the latent PR exceeds the strategy-ensemble PR by at least 1.69, for both players.

The overhead is featurisation residue (Section 4). Three independent experiments point the same way. The latent power spectrum breaks exactly at the input Fourier bandwidth, and under a K-sweep the break tracks K ($6 \to 6$, $12 \to 10$, $24 \to 22$, with 98% of latent power below K in every arm). The overhead is unrelated to the game's homotopy content across eight runs spanning 3–13 winding classes, including a control that varies sector count at matched capturability. And its variance splits 4.7 : 1 into parameter-initialisation and task-instance parts. Training *reduces* latent PR from ≈ 10.4 at initialisation. The game removes dimensions from the featurisation's default and stops where optimisation pressure ends, not at a game-set scale.

Topology appears as clusters, not dimensions (Section 5). Winding sectors are linearly cheap, for a simple reason: saddle strategies are constant headings, and constants span a 2-plane in control space no matter how many sectors they realise. The data agree. Within-class control families are essentially rank-1 (PR 1.10–1.22), and classes separate by $6\times$ in metric distance. So the game's homotopy content is real and prominent, carried by cluster geometry that no spectral statistic sees.

The one tempting spectral asymmetry does not survive its ablation (Section 7). Raw head spectra suggest a pursuer–evader asymmetry. A gauge lemma shows that raw head

spectra are not observables of the learned strategies at all, and the apparent concentration *inverts* under gauge fixing. What survives is a 2× pursuer excess of intermediate-band temporal power, stable across seeds. This is exactly the kind of signature one would happily attribute to the inf–sup structure, and a second-variation analysis even predicts its direction. We set in advance the condition that would rule that attribution out, ran the fully symmetrised protocol at four seeds, and report the result: the excess becomes seed-dependent (×1.25–1.99, straddling the pre-registered threshold, with sign inversions in one seed). We withhold the attribution. This negative result is one of the paper's more useful ones: gauge invariance plus seed stability, the usual bar for "real effect", was not enough.

Everything is trained under an *honesty contract*: no analytic value, no optimal controls, nothing in any loss that a solver could not observe. There is one exception, disclosed and measured (Section 2). On the torus, the standard distance-shaping regulariser is provably an affine image of the analytic value, and it supplies almost all of the evader's learning signal. We prove the identity, measure the gradient shares, and show with a from-scratch sweep that the ensemble-rank conclusions do not depend on it. An independently trained Hamilton–Jacobi–Isaacs critic checks the actor at the value level (Section 9). The critic's own convergence and failure analysis, including why its residual level cannot be read as quality, is a companion paper (Companion paper, 2026). A note on terminology. The *actor* is the whole self-play object, the coupled pursuer and evader head-banks; we name a single player when we mean one. The *critic* is the separately trained value network. This is not actor–critic in the reinforcement-learning sense: the critic never feeds gradients to the actor, and the two are trained independently. Proofs, the full audit trail, and all training artifacts are in the supplementary repository.

2 Game, prediction, architecture

The game. Pursuer P and evader E move at speeds $v_P = 1.2$, $v_E = 1$ on $\mathbb{T}^2 = [0, 2)^2$. Capture means $d_{\mathbb{T}}(x_E, x_P) \le \varepsilon = 0.08$ within horizon $t_{\max} = 4$, where $d_{\mathbb{T}}$ is the min-image distance. P minimises a capture-time functional and E maximises it. In relative displacement coordinates $\mathbf{\Delta} = x_E - x_P$ the feedback game has value $V^*(\mathbf{\Delta}) = (d_{\mathbb{T}}(\mathbf{\Delta}) - \varepsilon)_+/(v_P - v_E)$, attained by colinear play along the minimal displacement geodesic. This is the positional saddle of the Krasovskii–Subbotin theory (Krasovskii and Subbotin, 1988). Within reach $(v_P - v_E)\, t_{\max} + \varepsilon = 0.88$, which is 59.7% of the state space, capture is forced. Outside it the evader escapes the horizon, so on the full state space the value is the horizon-capped $V^* = \min\big(t_{\max},\, (d_{\mathbb{T}}(\mathbf{\Delta}) - \varepsilon)_+/(v_P - v_E)\big)$ of the companion critic (Companion paper, 2026), with the cap inactive on the reachable set the actor is scored on. We chose these constants as representative, not tuned. The value scale is $1/(v_P - v_E)$, and the dimensionless reach ratio $\rho = R/(L/2)$, with reach radius $R = (v_P - v_E)\, t_{\max} + \varepsilon$ (here $\rho = 0.88$), places the forced-capture region inside the domain but below the threshold $\rho = 1$ at which capturable play reaches the antipodal ridge, the cut locus of V^*. The one singular limit is $v_P \to v_E$, where the value diverges and the capturable set collapses onto the capture disk. Section 6 sweeps $v_P - v_E$ across that transition and finds the embedding dimensionalities unmoved by it. Two facts matter downstream. First, the saddle *open-loop* controls are constant headings $\hat{u}^*(t) \equiv \widehat{\mathbf{\Delta}}_0$: the optimal strategy manifold, as a subset of $L^2([0, t_{\max}]; \mathbb{R}^2)$, is the image of

a circle whose linear span is two-dimensional. A faithful embedding of optimal strategies should therefore have rank ≈ 2 plus soft-game corrections, *regardless of the game's homotopy content* (Proposition 1). Second, the open-loop game itself has no value: a committed evader concedes at most 1.11 (wrap-around interception caps the reachable-set advantage), while open-loop play against an informed maximiser concedes at least 1.66. So self-play converges, if anywhere, to a protocol-selected equilibrium, and we treat the trained object as exactly that.

Architecture and training. Following Tarancón-Álvarez et al. (2026), each player is a bank of $N_h = 16$ linear heads over a shared body. A tanh MLP maps Fourier features of time ($K = 12$ harmonics) to a latent curve $\boldsymbol{H}_\alpha(t) \in \mathbb{R}^{16}$, and head j, one per evader initial condition, outputs the unit heading $\hat{u}^j_\alpha(t) = W^j_\alpha \boldsymbol{H}_\alpha(t)/\|W^j_\alpha \boldsymbol{H}_\alpha(t)\|$. Trajectories are rolled out with a differentiable integrator. This gives a soft expected capture time $\widehat{T}$ (a differentiable surrogate for the first-capture time, averaged over rollout) and a terminal survival probability $\bar{S}$ per head. The players optimise

$$\mathcal{L}_P = \widehat{T} + \lambda_{\mathrm{nc}}\,\bar{S}\,t_{\max} + \lambda_d \langle d_{\mathbb{T}}(\boldsymbol{\Delta})\rangle, \qquad \mathcal{L}_E = -\widehat{T} - \lambda_d \langle d_{\mathbb{T}}(\boldsymbol{\Delta})\rangle, \tag{1}$$

where $\langle\cdot\rangle$ averages over the 16 heads and rollout time, λ_{nc} weights a no-capture penalty and λ_d the distance shaping. The shaping term uses the min-image distance $d_{\mathbb{T}}$, the quantity that enters V^*, not the raw Euclidean $\|\boldsymbol{\Delta}\|$ (the code wraps to the nearest image). Since $\widehat{T}$ is a time and $d_{\mathbb{T}}$ a length, $\lambda_d = 5$ is a dimensional coupling, not a dimensionless relative weight. The survival bonus ($\lambda_{\mathrm{nc}} = 2$) is *one-sided*, so the game as optimised is not exactly zero-sum. This is one of three protocol asymmetries, with the 3:1 pursuer:evader step ratio and the $10\times$ learning-rate ratio (Adam at $10^{-2}/10^{-3}$), that Section 7 returns to. Training runs 14,000 alternating iterations after a pursuer-only warm-up. All analytic gradients are verified against central finite differences to relative error $\leq 7 \times 10^{-9}$. No analytic value or control enters any loss. We drop the head-orthogonality penalty of Tarancón-Álvarez et al. (2026). There it fixes the latent gauge for a stable PCA. Here the strategy target is a known rank-two object (Proposition 1), the gauge is removed post hoc by Lemma 2, and the self-play equilibrium is non-stationary, so a basis-fixing penalty on a moving target would constrain the wrong thing.

The disclosed exception. On $\mathbb{T}^2$, off the capture set, $d_{\mathbb{T}}(\boldsymbol{\Delta}) = (v_P - v_E)\,V^*(\boldsymbol{\Delta}) + \varepsilon$ identically. So the shaping gradient *is* the value gradient up to a positive constant, and shaping is value-shaping on this geometry. (It is not on bounded geometries, where boundary play decouples distance from value.) Measured at the trained equilibrium, shaping supplies $\approx$100% of the evader's parameter-gradient norm and 6.4% of the pursuer's, and annealing $\lambda_d \to 0$ mid-training collapses the evader's gradient by a factor ≈ 45. The honest statement of the training signal is therefore: the pursuer is trained by the game, the evader largely by shaping. Every ensemble-rank claim below is checked against this dependence (Table 1, C5).

3 Two dimensionalities

Definitions. The *strategy ensemble* of player α is the indexed family $\mathcal{S}_\alpha = \{\hat{u}^j_\alpha(\cdot)\}_{j=1}^{16}$, the object the game theory constrains. We measure its linear dimension by the participation ratio $\mathrm{PR} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ of the temporal covariance spectrum of the pre-normalisation

	control	design	outcome
C7	warm-start collapse	pre-fit a rank-9.3 curved heading family (MSE 10^{-5}), then switch on the game loss	$\mathrm{PR}(Z_E)$ 6.5 $\to$ 2.5, $\mathrm{PR}(Z_P)$ 8.0 $\to$ 1.6 within 8k iterations: near-constancy is *selected*, not a default
C5	shaping sweep	$\lambda_d \in \{0, 1, 5\}$ from scratch (unshaped initialisation gradients 8.3/3.5: trainable)	$\mathrm{PR}(Z) = 1.6$–3.0 in all arms; rank robust; evader rank mildly shaping-dependent ($2.98 \to 1.94$)
C3	bandwidth sweep	input harmonics $K \in \{6, 12, 24\}$, all else fixed	spectral break at $k = 6, 10, 22$: tracks K; 98% of latent power below K in every arm
C4	sector restriction, capturability-matched	baseline radius multiset rotated into a $\pm 15^\circ$ diagonal sector; 4/16 capturable split preserved exactly	winding classes $12 \to 4$; $\mathrm{PR}(\boldsymbol{H}_E) = 5.22$, indistinguishable from the seed mean: no topology effect on the latent
C1	decoupled dispersion	8 init seeds at fixed ICs; 5 IC resamples at fixed init	$\mathrm{PR}(\boldsymbol{H}_E) = 5.4 \pm 0.9$ vs 5.94 ± 0.41: init variance dominates 4.7×
C2	symmetrised training	exact zero-sum loss (machine-checked), no warm-up, 1 : 1 steps, equal learning rates; 4 seeds × 5–6 snapshots	band-gap run means ×1.25–1.99, snapshot range $[0.66, 3.29]$: seed-dependent; inf–sup attribution withheld

Table 1: The attribution controls. For each we fixed a two-sided decision rule before running it, and outcomes are stated against those rules. C7 and C5 protect the strategy-rank claims, C3, C4 and C1 the featurisation attribution, C2 the asymmetry reading.

head outputs $Z^j = W^j \boldsymbol{H}(t)$, and of the unit-heading curves. We report both. The PR over 16 curves carries a bootstrap uncertainty of about half a unit, so ranges rather than decimals carry the conclusions. The *latent curve* $\boldsymbol{H}(t)$ is upstream, and the rank chain $\mathrm{rank}(Z) \le \min(\mathrm{rank}\, W_{\mathrm{stack}}, \mathrm{rank}_t\, \boldsymbol{H})$ makes $\mathrm{PR}(\boldsymbol{H})$ only an upper bound on strategy content: latent directions no head reads are invisible to play.

Result. Section 2 predicts $\mathrm{PR}(Z) \approx 2$ plus soft-game corrections. We measure $\mathrm{PR}(Z_E) = 2.99$ and $\mathrm{PR}(Z_P) = 1.91$ at the baseline. Across everything we ran (eight init seeds, five IC sets, the sector restrictions, the shaping sweep, the warm-start run, and twenty-one symmetric-protocol snapshots) the population range is [1.46, 2.99]. At the threshold used in the source construction, 90% of ensemble variance sits in 4 of 32 components (Z_E; 3 of 32 for Z_P), against Burgers' 3 of 20. So *the game's strategy family is as compressible as the PDE solution family*, not less. The latent curve, by contrast, holds $\mathrm{PR}(\boldsymbol{H}_E) = 5.4 \pm 0.9$ ($n = 8$ seeds; range $[3.65, 6.75]$), none of it read by the heads: the stacked, gauge-fixed head matrix has effective rank 2.5 (evader) and 1.9 (pursuer). The two populations do not overlap. The within-run form of the statement is sharpest: in all 43 checkpoints, for both players, $\mathrm{PR}(\boldsymbol{H})/\mathrm{PR}(Z) \ge 1.69$.

Rank two under stronger play. The trained actor captures only 23% of the capturable set (Section 9), so a sceptic could read its $\mathrm{PR}(Z) \approx 2$ as a property of *this* weak equilibrium rather than of the saddle Proposition 1 describes. It is not. On the capturable heads, where the pursuit strategy is actually determined, the unit-heading ensemble has PR ≈ 1.7 for

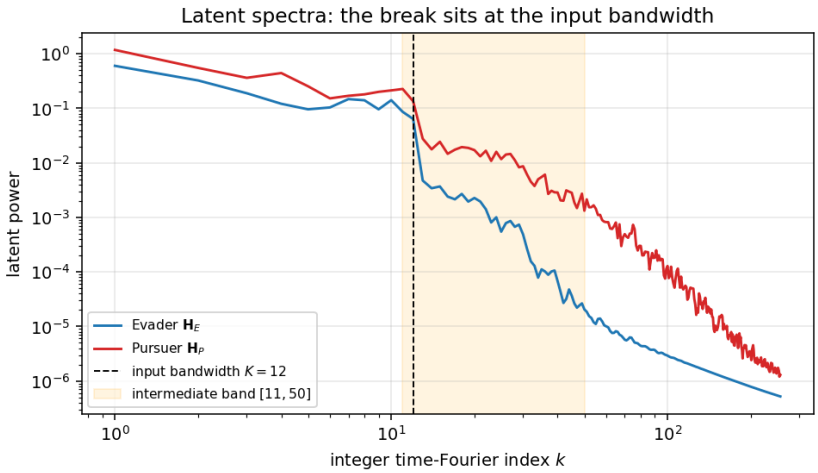


Figure 1: Integer-k latent power spectra. Both players' spectra are flat through the passband and collapse just above the input bandwidth $K = 12$ (dashed): between $k = 12$ and $k = 16$ the power drops by $\approx 27\times$ (evader) and $\approx 9\times$ (pursuer). Under the bandwidth sweep the break follows K ($6 \to 6$, $12 \to 10$, $24 \to 22$). The shaded band $k \in [11, 50]$ is the intermediate band used for the asymmetry statistics of Section 7.

the analytic constant-heading saddle, ≈ 2.1 for the mediocre actor, and ≈ 2.4–2.7 (three seeds) for the oracle-trained pursuer of Section 8, which captures the whole set. Optimal, near-optimal and mediocre pursuit are all rank two where it can be read. The PR ≈ 2 signature is a property of constant-heading play across strengths, not an accident of this equilibrium. (Over all 16 heads the oracle's rank inflates to ≈ 7, but only because its uncapturable heads chase evaders they cannot reach, so their headings are undetermined; this is an artifact of the free heads. The four capturable heads are the meaningful sample, small enough that these figures carry the same half-unit uncertainty.)

Selected, not default. Two controls rule out the readings a sceptic tries first. The architecture is not rank-limited: it fits an arbitrary curved unit-heading family of PR $= 9.3$ to MSE 10^{-5}. Started *from* that state, game training destroys the rank within 8,000 iterations (Table 1, C7) and converges onto the standard equilibrium. The shaped game objective actively selects near-constant strategies. And the selection is not an artifact of the value-equivalent shaping term: trained from scratch at $\lambda_d \in \{0, 1, 5\}$, the ensemble rank stays in $[1.6, 3.0]$ in every arm (C5), including the unshaped game, which trains despite having no value to converge to. What does move with λ_d, the evader's rank within that range, is reported as shaping-conditioned wherever it appears.

4 The latent overhead is featurisation residue

If the ≈ 5-dimensional latent overhead encoded game structure, it should respond to game quantities and be stable to architectural ones. The opposite holds on both counts.

The spectral break is architectural. The latent power spectrum (Figure 1) breaks at the input bandwidth, and the K-sweep (Table 1, C3) moves the break with K, not with any game scale. A tanh body driven by band-limited features populates frequencies above K only through harmonics, and that is what we see at every K.

It is unrelated to game topology. We ran eight runs spanning 3–13 realised winding classes: five seeds, two sector restrictions, and the capturability-matched diagonal-sector control C4, which cuts the class count from 12 to 4 while preserving the baseline's displacement-radius multiset and its 4/16 capturable split exactly. Across them $\mathrm{PR}(\boldsymbol{H}_E)$ shows no relation to class count: the full-diversity minimum (3.65) undercuts the three-class run (5.02), and C4 lands on the seed mean (5.22 against 5.4 ± 0.9, $n = 8$).

Its variance is a parameter-draw property. With initial conditions fixed, eight init seeds give $\mathrm{PR}(\boldsymbol{H}_E) = 5.4 \pm 0.9$. With the init fixed, five IC resamples give 5.94 ± 0.41, a $4.7 : 1$ ratio of variances (C1). The ratio comes from small samples ($n = 8$ and $n = 5$), so we read its direction (init dominates IC) and its order of magnitude, not the precise figure. And training *reduces* the latent rank from ≈ 10.4 at initialisation. The overhead is what optimisation pressure has not yet removed of the featurisation's default dimensionality, which is why it sits at the bandwidth and varies with the seed.

5 Topology: clusters, not dimensions

Proposition 1 (Constant curves are linearly cheap) *Saddle open-loop controls on $\mathbb{T}^2$ are constant headings; constant $\mathbb{R}^2$-valued curves span a two-dimensional subspace of $L^2([0, t_{\max}]; \mathbb{R}^2)$. Hence the linear dimension of the ideal strategy manifold is 2 independently of the number of homotopy classes its members realise, and no participation ratio can count winding sectors.*

Linear statistics cannot see winding sectors. Metric ones can. We label each head by the integer net winding of its evader trajectory, the rounded net displacement over $[0, t_{\max}]$, per axis. This displacement is the lift to the universal cover (the evader velocity integrated over the horizon, with no wrap applied), so w is a *signed* class in $\pi_1(\mathbb{T}^2) = \mathbb{Z}^2$, not a cyclic coordinate. Unit headings bound it by the reach $\|w\| \le v_E t_{\max}/L = 2$, and the extreme classes $w_x = \pm 2$ are opposite double-wraps, maximally separated in heading space ($L^2 = 2$) rather than identified across a periodic seam. The baseline's 16 trajectories realise 12 classes, distributed unevenly: one triple, two pairs, and nine singletons, so within-class comparison rests on only five pairs. Four of them are tight ($L^2 \le 0.07$, a $\gtrsim 20\times$ separation from the cross-class median 1.25) and near rank-one (PR 1.1–1.2). One same-winding pair is an outlier at 0.81, which lifts the *mean* within-class distance to 0.20 (a $6\times$ separation). So the typical within-class clustering is much tighter than the mean ratio suggests, with one exception, on the minority of heads that have a same-class neighbour at all. Homotopy content is therefore prominent in the ensemble, as *cluster geometry* (Figure 2), where it can be measured. This resolves what would otherwise look like a paradox in Section 3: a strategy family can be topologically rich and two-dimensional at once, and this one is.

Reflection control. The left–right structure of the winding cloud is a property of the initial-condition distribution, not of the winding coordinate. We reflect the one-sided sector across the y-axis ($x \to -x$) and retrain under an otherwise identical protocol: same

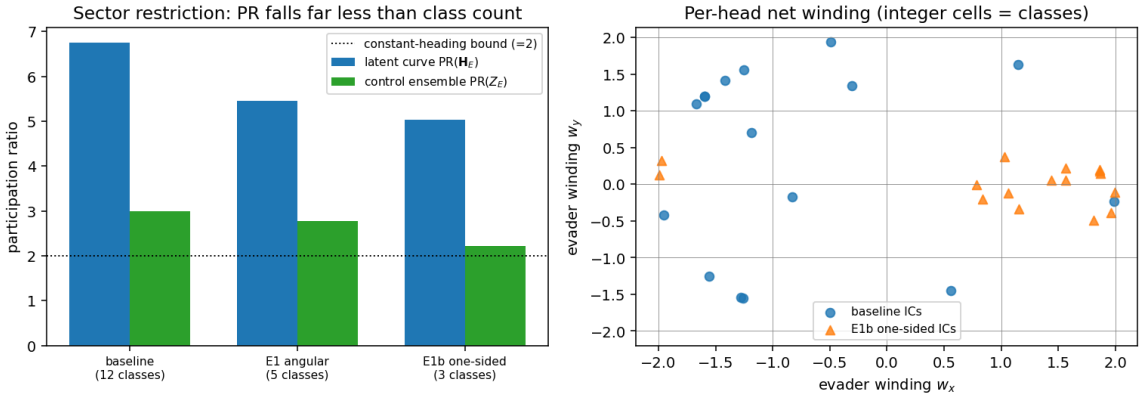


Figure 2: Sector experiments. Left: latent-curve PR (blue) vs control-ensemble PR (green) across the baseline and two sector-restricted runs; the dotted line is the constant-heading bound of Proposition 1. Right: per-head net winding $w \in \mathbb{R}^2$ for the baseline (12 integer classes, circles) and the one-sided restriction (3 classes, triangles). Restricting sectors collapses the *cluster count*; neither latent nor ensemble rank follows it.

initialisation, optimiser, and 14,000-iteration budget. This mirrors the cloud. The 14/16 heads that wound to $w_x > 0$ under the $+x$ sector wind to $w_x < 0$ under the $-x$ sector (mean w_x moving from $+1.05$ to -0.86), and the two antipodal wrap-arounds move from $w_x = -2$ to $w_x = +2$ (Figure 3). Because a reflection carries the cloud across $w_x = 0$, the winding is a genuine signed class in $\mathbb{Z}^2$. Were the axis cyclic, with $+2$ identified to -2, the reflection would be a null operation.

6 Parameter dependence

The constants of Section 2 are one point of a family, and we ask whether the embedding picture survives as we move through it. The controlling parameter is the speed difference $\Delta v = v_P - v_E$. It fixes the value scale $1/\Delta v$ and, through the reach radius $R = \Delta v\, t_{\max} + \varepsilon$, the reach ratio $\rho = R/(L/2)$. On the flat torus the capturable share of state space, $f(R)$, is the area fraction of $\{\boldsymbol{\Delta} : d_{\mathbb{T}}(\boldsymbol{\Delta}) \le R\}$. It is exactly computable and piecewise-smooth in the parameters, with a curvature break at $\rho = 1$, where the forced-capture disk first touches the antipodal cut locus, and saturation at $R = L/\sqrt{2}$ (Figure 4a). That break is the one structural transition in the family: below it every capturable trajectory is a short geodesic dash, above it capture wraps around the torus and the antipodal ridge enters the capturable set.

We retrain the actor from scratch at eight speed differences $\Delta v \in [0.05, 0.5]$, spanning $\rho \in [0.28, 2.08]$, three initialisation seeds each, with the architecture, horizon, geometry and evader IC set held fixed (the full 14,000-iteration budget; the $\Delta v = 0.2$ arm reproduces the qualitative split of Section 3). The result (Figure 4b) is a clean negative: neither dimensionality tracks the structural break. The strategy ensemble stays pinned near the

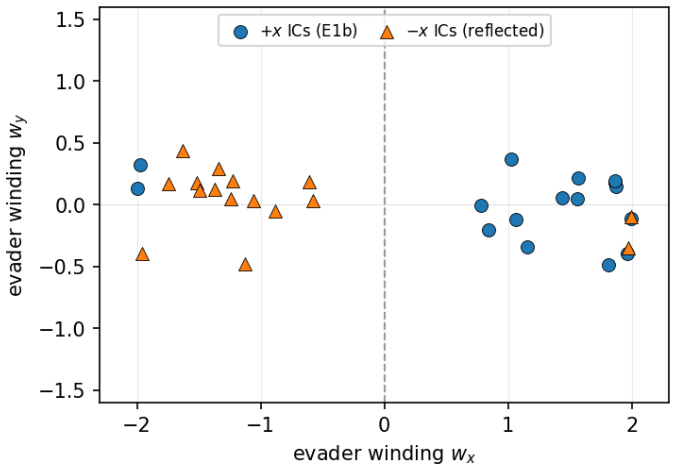


Figure 3: Reflection control. Per-head evader winding for the one-sided $+x$ sector (E1b, circles) and the same construction reflected to the $-x$ sector and retrained identically (triangles). The cloud mirrors about $w_x = 0$ (dashed line): the asymmetry tracks the deliberately one-sided initial conditions, and the signed winding does not wrap.

constant-heading bound. Its $\mathrm{PR}(Z_E) = 1.4$–3.1 across the whole family, marginally wider than the fixed-Δv runs' 1.5–3.0 because the sweep reaches into weaker- and stronger-game regimes, and it rises gently with Δv with no step at $\rho = 1$. This shows Proposition 1 is not an artifact of the baseline operating point: the strategies are two-dimensional wherever the game is capturable at all. The latent overhead stays at its featurisation scale ($\mathrm{PR}(\boldsymbol{H}_E) \approx 5$–$7$, no systematic trend in Δv), exactly as Section 4 predicts for a quantity set by the input bandwidth K rather than by any game scale. Capture performance, by contrast, rises with the capturable share as it must (Figure 4a). The game moves the actor's behaviour across the transition, but not the dimension of its representation.

7 The asymmetry that was not attributable

The trained players differ, and the differences invite game-theoretic readings. We take apart the two we found most tempting.

Lemma 2 (Gauge freedom of the heads) *The policy is invariant under per-head rescaling $W^j \to c_j W^j$, $c_j > 0$: unit headings depend only on the direction of $W^j \boldsymbol{H}$. Raw singular-value spectra of stacked head matrices are therefore not observables of the learned strategies.*

The raw head spectra differ by $3\times$ in scale between the players. By Lemma 2 this is gain, and gauge fixing inverts it (Figure 5). What survives gauge fixing is real: in the unit-heading ensembles the intermediate band $k \in [11, 50]$ carries 30.3% of the pursuer's temporal power against 15.8% of the evader's, a $\approx 2\times$ excess that persists when shaping is

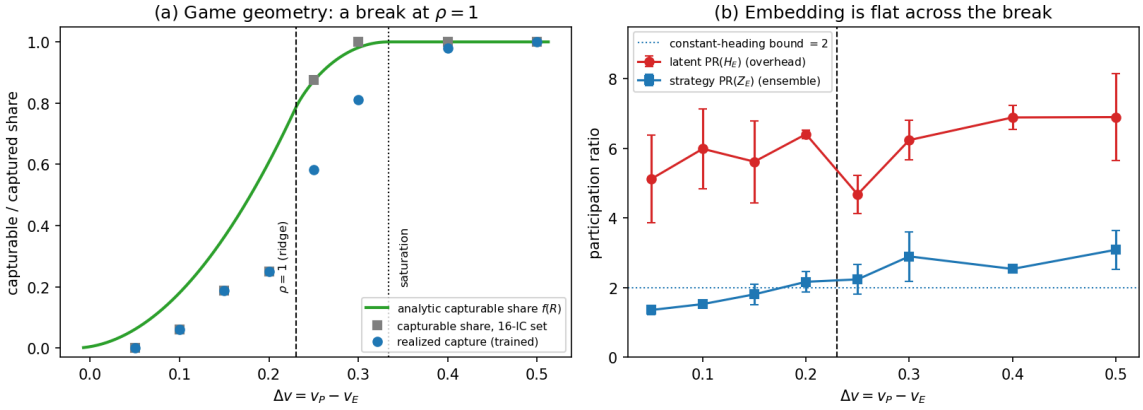


Figure 4: Embedding structure across the speed-difference family. (a) The analytic capturable share $f(R)$ on $\mathbb{T}^2$ is piecewise-smooth with a curvature break at $\rho = 1$ (dashed) and saturation at $R = L/\sqrt{2}$ (dotted); the fixed IC set's capturable share and the trained actor's realised capture track it. (b) Latent PR($\boldsymbol{H}_E$) and strategy-ensemble PR(Z_E) versus Δv (mean $\pm$ s.d., three seeds): the strategy rank stays at the constant-heading bound and the latent rank at its featurisation scale straight through the $\rho = 1$ break, so the representation does not see the game's structural transition.

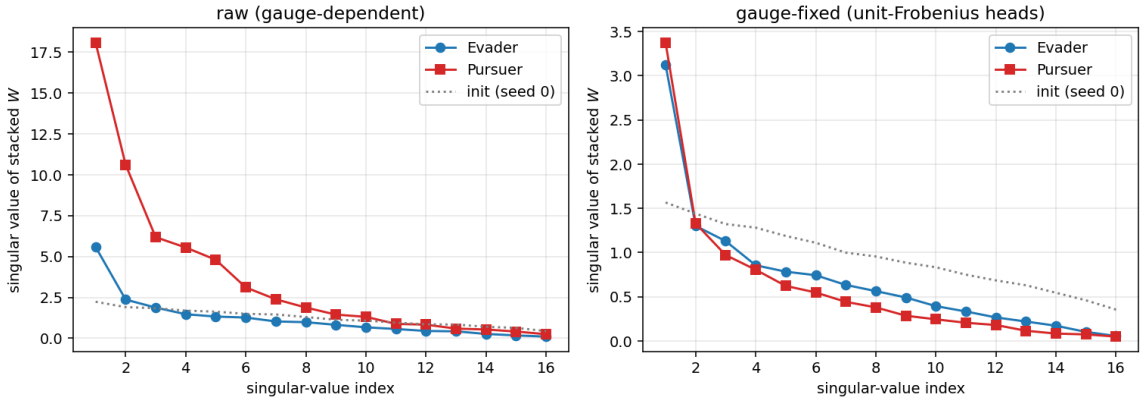


Figure 5: Head singular values, raw vs gauge-fixed. Left: raw spectra, pursuer max 18.1, evader max 5.6. Right: after fixing each head to unit Frobenius norm the apparent concentration *inverts* (effective rank 2.53 evader vs 1.91 pursuer). The raw contrast tracks weight motion (4.1× vs 1.3× the initialisation norm), so it is optimisation gain, not strategy structure.

annealed away (22.7% vs 11.5%) and is reproducible across seeds under the baseline protocol (×1.44–1.92). There is even theory pointing the right way. At the saddle, the second variation of capture time weights pursuer heading perturbations by v_P/v_E and adds an

ε^{-1}-concentrated endgame term that binds only on capturing trajectories. The minimising player keeps receiving broadband curvature after the maximising player's objective has saturated (soft no-capture mass 0.70 at equilibrium).

But the trained asymmetry is confounded: the protocol itself treats the players asymmetrically (one-sided survival bonus, 3:1 steps, 10× learning-rate ratio). We fixed a two-sided rule for the deciding experiment: gap persists $\geq 1.5\times$ under a fully symmetrised protocol $\Rightarrow$ attributable to the inf–sup structure; collapse $\Rightarrow$ protocol artifact. We then ran exact zero-sum training (loss symmetry machine-checked each step, no warm-up, 1:1 steps, equal learning rates) at four seeds, five to six snapshots each, the equilibrium being oscillatory as expected for simultaneous gradient play (Mescheder et al., 2018; Daskalakis et al., 2018). The outcome (Table 1, C2): run-mean gaps of $\times 1.25$, 1.51, 1.64, 1.99, straddling the threshold, with snapshot range $[0.66, 3.29]$ and sign inversions in one seed, where the asymmetric protocol was seed-stable. The rule returns no verdict. The raw-scale gap, meanwhile, equalises exactly as the gauge analysis predicts ($\sigma_{\max}$ 12.5/12.5). We therefore withhold the attribution: the asymmetry is at minimum strongly protocol-shaped, and deciding whether any of it is game-theoretic needs replication power this study does not have. The methodological point stands regardless. A gauge-invariant, seed-stable, theory-consistent signature was still protocol-sensitive, and only a pre-registered ablation could say so.

8 The equilibrium is protocol-limited, not capacity-limited

The trained actor is a genuine but mediocre equilibrium (it captures a minority of the capturable set, Section 9), and adversarial self-play is known to be unstable: simultaneous gradient play in a zero-sum game rotates around the equilibrium rather than converging to it (Mescheder et al., 2018; Daskalakis et al., 2018), the two-dimensional analogue of the limit cycles of rock–paper–scissors. Is the mediocrity a limit of the open-loop *networks* or of the co-adaptive *protocol*? We separate the two by re-training the pursuer under four protocols at matched budget, initial conditions, architecture and seeds (three each), scoring the mean pursuit gap on the analytically capturable heads and its late-training oscillation (Figure 6).

Against a *fixed* optimal evader, the constant-heading saddle of Proposition 1, an oracle opponent that removes co-adaptation entirely, the pursuer converges monotonically to full capture (gap 1.4×10^{-3}, oscillation s.d. 5×10^{-4}). A single open-loop head per initial condition has ample capacity to realise optimal pursuit, so representation is not the bottleneck. The instability lives in the coupling. Strict simultaneous (1:1) play is oscillatory and seed-dependent (capture 0.75–1.0, oscillation up to 0.04, one seed losing a head), the same signature the symmetrised control of Section 7 showed on the spectral side. Undamped alternating best response (train one player toward convergence, switch, repeat) *cycles*: pursuer and evader alternately win outright (capture 0 at the measured phase, oscillation 0.17, an order of magnitude above every other arm), never settling. This is exactly the best-response cycle that only time-averaging (fictitious play) is known to close. The baseline's 3:1 pursuer:evader step ratio is the steadiest of the co-adaptive arms, its asymmetry damping the cycle. The equilibrium's quality is therefore a property of the training dynamics, not of the representation. That is the limit of the "mediocre" verdict, and the reason the co-training plan of Section 11 must specify an opponent schedule, not just a larger network. (The oracle opponent uses the analytic optimal strategy, so it is a

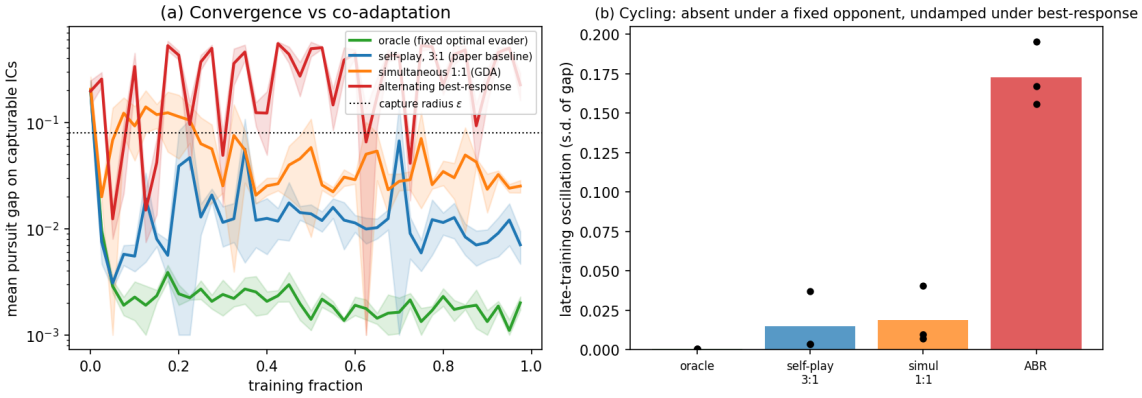


Figure 6: Actor training dynamics under four protocols (matched budget, ICs, architecture; three seeds). (a) Mean pursuit gap on the capturable heads vs training fraction (band: s.d. over seeds): the oracle (fixed optimal evader) descends monotonically below the capture radius; alternating best-response sawtooths between win and loss; simultaneous 1:1 oscillates; the 3:1 baseline is damped. (b) Late-training oscillation amplitude grows monotonically with the symmetry of the co-adaptation, from zero for the fixed opponent to largest for alternating best-response. Capacity is constant across arms; only the coupling changes.

diagnostic baseline, outside the honesty contract, not a training method we advocate. We test only the pursuer this way. The mirror arm, an evader against a fixed optimal pursuer, is degenerate by construction, since a slower evader has no escaping strategy against optimal faster pursuit, only delay.)

Can an honest protocol recover the oracle's stability? The oracle removes co-adaptation by fixing the opponent to the analytic optimum, which the honesty contract forbids. We asked whether a *self-play* correction can recover its stability. We tested the standard game-optimisation fixes that provably tame cycling in convex–concave games (extragradient, optimistic gradient descent–ascent (Daskalakis et al., 2018), and iterate averaging) on the identical 1:1 setting, three seeds, at matched budget. None closes the gap (Figure 7). Cesàro parameter-averaging is actively *destructive* (capture collapses to 0.4): the iterates orbit a limit cycle rather than a point, and their mean is a blurred, non-capturing policy. Optimism and extragradient are inconsistent across seeds (capture 0.58–0.83, oscillation still 10–100× the oracle's). We also tried the combination the nonconvex min–max literature favours, optimistic GDA with an exponential moving average of the iterate. It attains the *lowest* oscillation of any honest arm, but only by trading capture down to 0.58, so it too stays out of the basin. (Forcing the smaller *symmetric* step the convergence theory assumes makes it worse: the pursuer needs its larger step to learn pursuit within budget, and capture collapses. The step asymmetry is load-bearing, not merely a cycle-damper.) Across the toolkit the honest methods trace a capture–stability trade-off, and none enters the oracle basin (Figure 7a). These methods are not wrong. Their guarantees are for convex–concave (or bilinear) games, while this min–max problem is nonconvex–nonconcave,

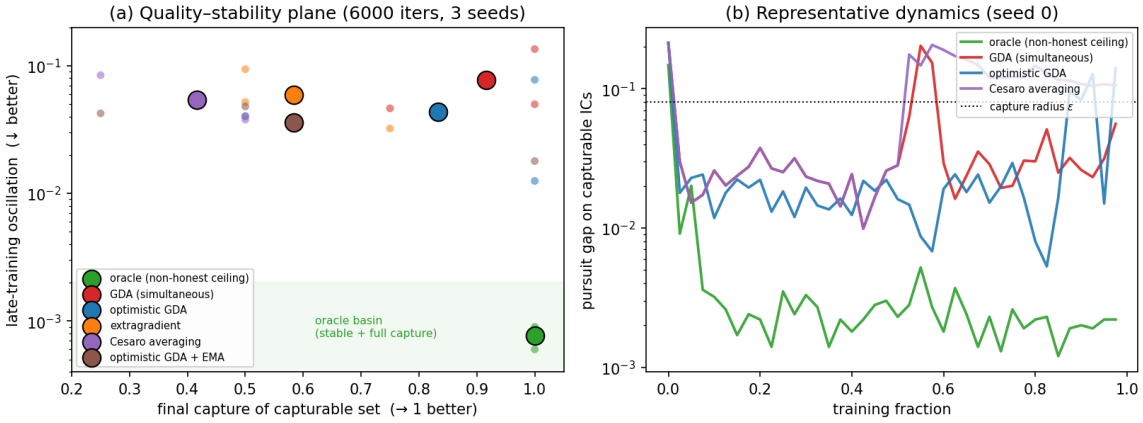


Figure 7: Honest self-play corrections do not reach the oracle (6,000 iterations, three seeds; consistent with a 12,000-iteration check). (a) Quality–stability plane: final capture (right is better) versus late-training oscillation (down is better). The oracle occupies a stable, full-capture basin (shaded) that no honest correction enters (simultaneous GDA, optimistic GDA, extragradient, Cesàro averaging, or optimistic GDA with iterate EMA); large markers are protocol means, small markers seeds. (b) Representative dynamics: the oracle converges and stays; GDA oscillates; the averaged model sits high (its mean policy does not capture).

so the corrections do not carry over directly. Closing the honest-versus-oracle gap, a self-play protocol that reaches the oracle basin without the oracle, is a concrete open problem, and the quality–stability plane of Figure 7a is the yardstick we propose for it.

Locating the gap with Shapley values. We cannot close the gap, but we can say precisely where it lives. Treating disjoint frequency bands of the pursuer's time input as players, we compute the exact Shapley attribution of capture (against the fixed optimal evader, so both pursuers face the same task) for an oracle-trained and a self-play-trained pursuer (Figure 8). The oracle builds capture almost entirely from a *low-frequency* heading modulation: the $k \in [1, 4]$ band carries 68% of its capture, and the linear, mid and high bands little. The self-play pursuer never acquires that band. Every band's Shapley value is ≈ 0, and against the optimal evader it falls back on its constant-heading (DC) response, capturing 0.17 against the oracle's 1.00 ($n = 3$ seeds). The result has two parts. As a diagnosis, the co-adaptation gap is a *specific missing component*, the low-frequency pursuit correction, not a diffuse deficit. Shapley localises it without touching a weight, which is exactly its scope: an attribution of a trained model, not a training method. As independent corroboration of Section 3, capture is carried by near-constant, low-frequency headings, while the higher-frequency content (the latent overhead) contributes essentially nothing to play, now confirmed by a Shapley analysis rather than a spectral one.

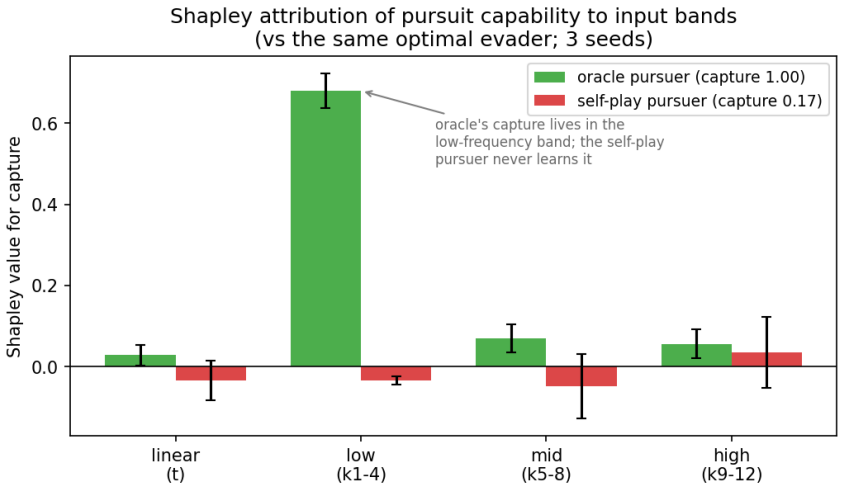


Figure 8: Exact grouped-Shapley attribution of capture to frequency bands of the pursuer's time input, evaluated against the same optimal evader (three seeds; masked features set to their time-mean). The oracle pursuer's capture concentrates in the low-frequency band; the self-play pursuer acquires no useful band and reverts to its constant-heading response. Shapley locates the oracle gap (a missing low-frequency modulation) but, computed on trained networks, does not close it.

9 Value-level verification

A representation audit of a game-trained network needs an independent standard of play. We train a critic on the HJI residual of the displacement-form game plus its two physical boundary conditions, again nothing analytic, and use it three ways. This verification critic runs at the actor's horizon $t_{\max} = 4$ ($\rho = 0.88$), so the capturability normalisation below is matched to the actor's reachable set. The companion paper's longer-horizon torus arm ($\tau_{\max} = 8$, $\rho = 1.7$) probes the ridge-active regime separately.

Performance, normalised by capturability. The critic's feedback policy captures 31% of the analytically capturable test set; the open-loop actor captures 23%. (Unnormalised grid fractions, 19% and 14%, understate both, since only 59.7% of the grid is capturable at all.) The actor is a real equilibrium of middling strength, which bounds the scope of every conclusion above.

The directional comparison factorises. Along rollouts, the measured actor–critic alignment equals the product of each side's alignment with the analytic saddle direction $\widehat{\boldsymbol{\Delta}}$ to within 0.03 ($0.882 \times 0.484 = 0.427$ vs 0.433 measured; $0.882 \times 0.576 = 0.508$ vs 0.541 with a stronger critic). So there is no actor–critic coupling beyond what the saddle direction mediates, and the critic's gradient field is the bottleneck. The companion paper traces that bottleneck to the critic's parameterisation rather than to residual training itself (Companion paper, 2026).

Topology-level consistency is unpopulated at this equilibrium. The companion paper's structured critic exposes a homotopy selector validated exactly against the analytic pursuit chart. This enables a topology-level actor–critic consistency test: do the actor's winding

classes agree with the critic's active chart along rollouts? Run on this actor, the test returns a well-defined empty verdict. The only heads whose relative displacement ever contests the antipodal ridge are the escaping, horizon-limited ones, exactly where the selector is analytically degenerate, while every capturable head stays clear of it. Strategy topology and value topology do not interact at the open-loop equilibrium. This is one more face of the actor's early commitment, and the concrete observable that the co-training plan of Section 11 aims to populate.

A null for the latent bridge. The natural conjecture that the latent overhead encodes value information fails its control. The critic's value along rollouts is decoded from the top two latent components at cross-validated $R^2 = 0.46$, but from *time alone* at 0.53. The leading latent components are smooth functions of t, so the decodability is not latent-specific. We report this null with the same standing as the positive results, and it closes the last route back to "the overhead is game content".

10 Related work

Neural approaches to pursuit–evasion and differential games mostly *solve* the game: multi-agent RL and self-play (Lowe et al., 2017; Baker et al., 2020), value-function and reachability methods (Bansal and Tomlin, 2021). Our question is different. We ask what the trained representation contains, and we are not aware of prior representation audits of game-trained physics-informed architectures. The construction and the latent-PCA method are from Tarancón-Álvarez et al. (2026), and the Burgers comparison is protocol-matched to that source (its variance thresholds and standardisation conventions). The training-signal disclosure connects to the known sensitivity of self-play equilibria to optimisation asymmetries (Mescheder et al., 2018; Daskalakis et al., 2018). The critic builds on physics-informed residual training (Raissi et al., 2019; Karniadakis et al., 2021). Its failure modes, and the standards under which any critic-derived number here should be read, are developed in the companion paper (Companion paper, 2026).

11 Limitations and outlook

Ground truth here comes at a cost in scope. All embedding conclusions concern one game family on one geometry, at one equilibrium of an open-loop actor that captures 23% of the capturable set, and the PDE-side comparison rests on a single source family. Three extensions would move the conclusions rather than tighten them. A *closed-loop, state-conditioned actor* is the only one that could alter the ensemble-rank story: feedback strategies are not constant headings, so Proposition 1 no longer bounds them. *Bounded geometries* (disk, ball, Möbius strip) make the shaping term harmless, since distance is provably not value-equivalent there, which makes them the clean venue for repeating the embedding analysis. And *honest co-training* against the companion paper's structured critic, whose validated homotopy selector meets this paper's winding analysis at the topology level, is the planned route to a value–representation bridge that our nulls show does not yet exist. A fourth extension is now open rather than closed. Section 6 varies the speed difference but holds ε, $t_{\max}$ and L fixed, and the clean piecewise structure of the game geometry (Figure 4a) invites promoting all four to *conditioning inputs* of a single embedding. That would map how the

latent and strategy spectra deform across the phase diagram. From the piecewise-linearity of V^* in these parameters, we expect (and pre-register) smooth dependence broken only at the capturability ($v_P \to v_E$) and ridge-activation ($\rho = 1$) transitions. A fifth extension is more open-ended and follows from Section 8. The honest–versus–oracle stability gap left standing there motivates a *purpose-built nonconvex–nonconcave min–max optimiser*, an asymmetry-preserving optimistic method with adaptive per-player preconditioning and recency-weighted iterate averaging, aimed at reaching the oracle basin under pure self-play. The concrete first attempt we would make is *neural fictitious self-play* with an opponent pool ("league" training), which averages the *play* rather than the weights. This is the version of averaging with zero-sum convergence guarantees, and the one our parameter-averaging arm was a degenerate substitute for. It exposes the pursuer to strong, self-generated opponents instead of only the current weak one. This is a study in its own right, so we pose it here, with the quality–stability plane (Figure 7a) as its benchmark, rather than attempt it.

The transferable method is smaller than the paper and easy to state: define the embedded object before measuring it; predict its dimension from theory; give every interpretive claim a pre-registered, two-sided control; and let gauge lemmas and nulls kill readings early. On this game, that discipline turned a tempting headline ("strategic behaviour is seven-dimensional") into a sharper, better-supported one. The strategies are exactly as simple as the game theory says, the extra dimensions belong to the featurisation, and the topology lives in the metric, where the spectral statistics could not see it.

Reproducibility statement

All training code (a from-scratch NumPy pipeline with every analytic gradient verified against central finite differences to $\leq 7 \times 10^{-9}$), checkpoints for all 43 runs and snapshots, the analysis suite that regenerates every number and figure, and the complete audit trail (pre-registered designs, decision rules, and the per-control analysis records cited in Table 1) are versioned in the supplementary repository. No result in this paper depends on unreleased data or manual steps.

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